Certificate of Analysis

Sample Name:	LIT CULTURE BUTTERSCOTCH
Client:	Free River Distributing
Sample Code:	DTS-260804-084
Matrix Name:	Beverage & Liquid - Shot
Type / Result:	Batch Release - Pass

cambium analytica
an NSF company

Received Date:	Thu, Aug 6, 2026
Published Date:	Tue, Aug 11, 2026
Batch/Lot Code:	BTRSCH-045
Batch Size:	N/A
Sample Size:	1U
Average Unit Weight:	15.9135g (Density (g/mL) * 15mL package. 3 servings/package)

RESULT SUMMARY

Mitragynine	40.95 mg /serv
Total Major Alkaloids	48.80 mg /serv

Code	Test	Method
ALKU	Kratom Alkaloids	High Level
ALKL	Kratom Alkaloids	Low Level
SAL	Salmonella spp.	qPCR
ECOLI	Total Coliforms & E. coli	Plate
TAMC	Total Aerobic Bacteria	Plate
DEN	Density of Liquids	
HVMET	Heavy Metals	Big 4
TYMFD	Total Yeast & Mold	Plate
SOLHM	Residual Solvents	National Panel
7OHFL	7–Hydroxymitragynine	Florida

Approvals

RESULTS REVIEWED BY:

Leslie Nagy
Laboratory Director
Cambium Analytica
Tuesday, Aug 11, 2026

RESULTS CERTIFIED BY:

Douglas Smith
VP - Scientific Operations
Cambium Analytica
Tuesday, Aug 11, 2026

Lab Information

Address: 1230 Woodmere Ave, Traverse City, MI 49686 **Phone:** 231.252.3669 **Accreditation:** ISO/IEC 17025:2017 – #108157

ALKU Kratom Alkaloids - High Level

LAB-TM-052 - Determination of Kratom Alkaloid Content by UPLC-DAD
ALKU-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
Mitragynine	0.7720 %	7.7198 mg/g	40.95 mg	122.85 mg	N/A	0.0507 mg/g	0.1013 mg/g	N/A
Paynantheine	0.0599 %	0.5992 mg/g	3.18 mg	9.54 mg	N/A	0.0507 mg/g	0.1013 mg/g	N/A
Speciogynine	0.0444 %	0.4440 mg/g	2.35 mg	7.06 mg	N/A	0.0507 mg/g	0.1013 mg/g	N/A
Speciociliatine	0.0437 %	0.4367 mg/g	2.32 mg	6.95 mg	N/A	0.0507 mg/g	0.1013 mg/g	N/A
Total Major Alkaloids *	0.9200 %	9.1996 mg/g	48.80 mg	146.40 mg	N/A	N/A	N/A	N/A

**Total Major Alkaloids is calculated as the sum of Mitragynine, Paynantheine, Speciociliatine and Speciogynine.*

ALKL Kratom Alkaloids - Low Level

LAB-TM-047 - Determination of Kratom Alkaloid Content by LC-TQ
ALKL-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
7-Hydroxymitragynine	0.00016 %	0.00165 mg/g	0.01 mg	0.03 mg	N/A	0.002 ug/g	0.011 ug/g	N/A
Mitraphylline	ND	N/A	N/A	N/A	N/A	0.004 ug/g	0.019 ug/g	N/A
Total Minor Alkaloids *	0.00016 %	0.00165 mg/g	0.01 mg	0.03 mg	N/A	N/A	N/A	N/A

**Total Minor Alkaloids is calculated as the sum of 7-Hydroxymitragynine and Mitraphylline.*

SAL Salmonella spp. - qPCR - 25 g

LAB-TM-063 - Qualitative Detection of Presumptive Salmonella spp. in Foods and Dietary Supplements
SAL-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Salmonella spp.	ND /25 g	N/A	N/A	N/A	N/A

USP <62> - Vendor Modified, USP <2022> - Vendor Modified, USP <1223>, AOAC PTM 121802, AOAC OMA 2020.02

ECOLI Total Coliforms & E. coli - Plate - 25 g - Full Range

LAB-TM-059 - Enumeration of Escherichia coli and Total Coliform in Foods and Dietary Supplements
ECOLI-DTS-260804-084-01 - TUE, AUG 11, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
E. coli	ND	N/A	10 CFU/g	10 CFU/g	N/A
Total Coliforms	ND	N/A	10 CFU/g	10 CFU/g	N/A

USP <61> - Vendor Modified, USP <2021> - Vendor Modified, USP <1223>, AOAC PTM 051801, AOAC OMA 2018.13

Testing was conducted on a 25.3094 gram aliquot

TAMC — Total Aerobic Bacteria - Plate - 25 g - Full Range

LAB-TM-060 - Enumeration of Total Aerobic Count in Foods and Dietary Supplements
TAMC-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Total Aerobic Count	10 CFU/g	N/A	10 CFU/g	10 CFU/g	N/A

USP <61> - Vendor Modified, USP <1223>, AOAC PTM 121403, AOAC OMA 2015.13

Testing was conducted on a 25.3094 gram aliquot

DEN — Density of Liquids

LAB-TM-017 - Brix & Density Analysis
DEN-DTS-260804-084-01 - FRI, AUG 7, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Density	1.0609 g/mL	N/A	N/A	N/A	N/A
Specific Gravity *	1.0628	N/A	N/A	N/A	N/A

**Specific gravity is calculated using the density of water at 20 °C (0.9982 g/mL) using the equation: [Specific Gravity = (Density of sample in g/mL) ÷ 0.9982 g/mL]*

HVMET — Heavy Metals - Big 4

LAB-TM-044 - Determination of Heavy Metals by ICP-MS
HVMET-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Value (mg/g)	Per Serving	Per Package	Action Limit	LOD	LOQ	Status
Arsenic	ND	N/A	N/A	N/A	N/A	9.255 ug/kg	25.707 ug/kg	N/A
Cadmium	ND	N/A	N/A	N/A	N/A	1.851 ug/kg	6.427 ug/kg	N/A
Lead	ND	N/A	N/A	N/A	N/A	1.131 ug/kg	6.427 ug/kg	N/A
Mercury	ND	N/A	N/A	N/A	N/A	0.360 ug/kg	0.643 ug/kg	N/A

USP <233>, USP <730>, USP <1730>

TYMFD — Total Yeast & Mold - Plate - 25 g - Full Range

LAB-TM-061 - Enumeration of Yeast and Mold in Foods and Dietary Supplements
TYMFD-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Total Mold	ND	N/A	10 CFU/g	10 CFU/g	N/A
Total Yeast	ND	N/A	10 CFU/g	10 CFU/g	N/A
Total Yeast and Mold *	0 CFU/g	N/A	N/A	N/A	N/A

**Total Yeast and Mold is calculated as the sum of Total Yeast and Total Mold*

USP <61> - Vendor Modified, USP <1223>, AOAC PTM 121301, AOAC OMA 2014.05

Testing was conducted on a 25.3094 gram aliquot

SOLHM

Residual Solvents - National Panel

ANA-TM-053 - Determination of Residual Solvents
SOLHM-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
1,1-Dichloroethene	ND	N/A	5 ug/g	10 ug/g	N/A
1,2-Dichloroethane	ND	N/A	0.5 ug/g	1.00 ug/g	N/A
2-Methylbutane	ND	N/A	5 ug/g	10 ug/g	N/A
2-Methylpentane	ND	N/A	5 ug/g	10 ug/g	N/A
2,2-Dimethylbutane	ND	N/A	5 ug/g	10 ug/g	N/A
2,3-Dimethylbutane	ND	N/A	5 ug/g	10 ug/g	N/A
3-Methylpentane	ND	N/A	5 ug/g	10 ug/g	N/A
Acetone	ND	N/A	5 ug/g	10 ug/g	N/A
Acetonitrile	ND	N/A	5 ug/g	10 ug/g	N/A
Benzene	ND	N/A	0.18 ug/g	0.50 ug/g	N/A
Butane	ND	N/A	5 ug/g	10 ug/g	N/A
Chloroform	ND	N/A	0.78 ug/g	1 ug/g	N/A
Ethanol	<LOQ	N/A	5 ug/g	10 ug/g	N/A
Ethyl Acetate	ND	N/A	5 ug/g	10 ug/g	N/A
Ethyl Ether	ND	N/A	5 ug/g	10 ug/g	N/A
Ethylene Oxide	ND	N/A	2 ug/g	4 ug/g	N/A
Heptane	ND	N/A	5 ug/g	10 ug/g	N/A
Hexane	ND	N/A	5 ug/g	10 ug/g	N/A
Isobutane	ND	N/A	5 ug/g	10 ug/g	N/A
Isopropyl Alcohol	ND	N/A	5 ug/g	10 ug/g	N/A
Methanol	ND	N/A	5 ug/g	10 ug/g	N/A
Methylene Chloride	ND	N/A	5 ug/g	10 ug/g	N/A
Neopentane	ND	N/A	5 ug/g	10 ug/g	N/A
Pentane	ND	N/A	5 ug/g	10 ug/g	N/A
Propane	ND	N/A	5 ug/g	10 ug/g	N/A
Toluene	ND	N/A	5 ug/g	10 ug/g	N/A
Total Xylenes	ND	N/A	5 ug/g	10 ug/g	N/A
Trichloroethylene	ND	N/A	0.5 ug/g	1 ug/g	N/A
Total Butanes *	0.000 ug/g	N/A	N/A	N/A	N/A
Total Hexanes *	0.000 ug/g	N/A	N/A	N/A	N/A
Total Pentanes *	0.000 ug/g	N/A	N/A	N/A	N/A

Total Butanes is calculated as the sum of Butane and Isobutane
Total Hexanes is calculated as the sum of Hexane, 2,2-Dimethylbutane, 2,3-Dimethylbutane, 2-Methylpentane, and 3-Methylpentane.
Total Pentanes is calculated as the sum of Pentane, 2-Methylbutane, and Neopentane.

Modified USP <467>

7OHFL

7–Hydroxymitragynine - Florida - Dry–Weight Basis

7–Hydroxymitragynine - (Florida — Dry–Weight Basis)

7OHFL-DTS-260804-084-01 - MON, AUG 10, 2026

Analyte	Value	Action Limit	LOD	LOQ	Status
Moisture (%) by Input *	65.000 %	N/A	N/A	N/A	N/A
7-Hydroxymitragynine - As Received *	1.65 ppm	N/A	N/A	N/A	N/A
7-Hydroxymitragynine - Dry-Weight *	4.71 ppm	400 ppm	N/A	N/A	PASS

*The Moisture (%) by Input result may represent either a client-provided input or a value measured by Cambium. This value is incorporated into associated calculated analytes for reporting purposes.

*7-Hydroxymitragynine – As Received: The "As Received" value represents the concentration of 7-hydroxymitragynine, expressed in parts per million (ppm), as reported in the Kratom Alkaloids – Low Level test. This corresponds to the measured percentage of 7-hydroxymitragynine in the sample as received.

*7-Hydroxymitragynine (Dry-Weight Basis): The dry weight concentration of 7-hydroxymitragynine is calculated using the following formula: Dry Weight 7-Hydroxymitragynine (ppm) = 7-Hydroxymitragynine (ppm) x (100 / (100 – Moisture by Input (%)))